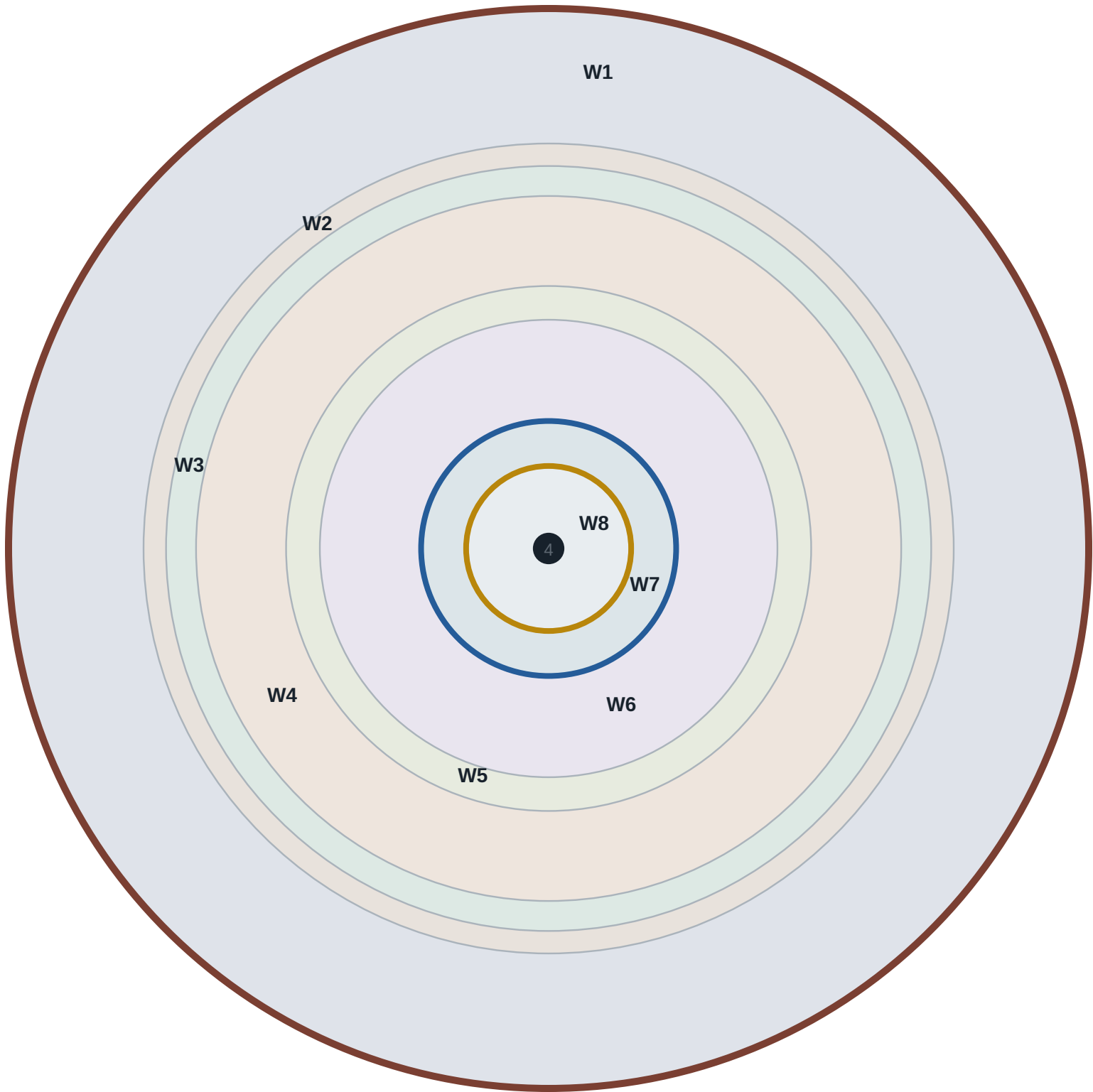


The Spindle of Necessity — Plato's Eight Whorls

Deterministic reconstruction from Plato's ordinal ranking and Michell's musical-number interpretation



Exact radial geometry: shaft $r=4$; outer $r=144$; scale $\times 180$
Exact arithmetic reproduction validates the internal numerical chain of Michell's printed construction.
It does not establish that Plato intended Michell's numerical assignment.

Provenance boundary

- PLATO: ordinal rim-width order only
- MICHELL: musical numbers, shaft rule, and scale $\times 180$
- PROJECT: exact cumulative reconstruction; 0 free parameters

Whorl construction

Layer	Whorl	Width	Cum. radius	$\times 180$
Shaft	—	—	4	720
Whorl 8	8	18	22	3960
Whorl 7	7	12	34	6120
Whorl 6	6	27	61	10980
Whorl 5	5	9	70	12600
Whorl 4	4	24	94	16920
Whorl 3	3	8	102	18360
Whorl 2	2	6	108	19440
Whorl 1	1	36	144	25920

Deterministic chain

Michell musical sequence: 6, 8, 9, 12, 18, 24, 27, 36

Plato = Michell width order: 1, 6, 4, 8, 7, 5, 3, 2

Centre-out widths: 18, 12, 27, 9, 24, 8, 6, 36

Cumulative radii: 4, 22, 34, 61, 70, 94, 102, 108, 144

Key highlighted boundaries

- Earth-radius / Magnesia correspondence: $r=22 \rightarrow 3960$
- New Jerusalem outer-circle correspondence: $r=34 \rightarrow 6120$
- Michell precessional correspondence: $r=144 \rightarrow 25920$